Results: nested subject-wise CV (1-s windows, 75 % overlap)

Outer 4-fold × inner 3-fold cross-validation, split by subject. Hyper-parameters and the number of epochs are chosen inside each outer fold using only its training subjects; the final model is retrained on all outer-training subjects and tested once on the held-out outer-test subjects. Each subject receives exactly one outer-test prediction (mean of its 1,197 segment probabilities, threshold 0.5), and all metrics are computed once on these pooled predictions. Positive class = more severe group.

Pooled subject-level results

Task	Model	ACC	Bal. ACC	F1	Precision	Recall	Specificity	AUC	TN / FP / FN / TP
None vs (Mild + Severe)	EEGNet	0.581	0.573	0.519	0.528	0.509	0.638	0.576	44 / 25 / 27 / 28
None vs (Mild + Severe)	ShallowFBCSPNet	0.605	0.595	0.533	0.560	0.509	0.681	0.582	47 / 22 / 27 / 28
None vs (Mild + Severe)	Deep4Net	0.548	0.533	0.440	0.489	0.400	0.667	0.530	46 / 23 / 33 / 22
(None + Mild) vs Severe	EEGNet	0.597	0.378	0.038	0.032	0.048	0.709	0.385	73 / 30 / 20 / 1
(None + Mild) vs Severe	ShallowFBCSPNet	0.621	0.469	0.175	0.139	0.238	0.699	0.421	72 / 31 / 16 / 5
(None + Mild) vs Severe	Deep4Net	0.556	0.354	0.035	0.028	0.048	0.660	0.408	68 / 35 / 20 / 1
Mild vs Severe	EEGNet	0.600	0.613	0.560	0.483	0.667	0.559	0.581	19 / 15 / 7 / 14
Mild vs Severe	ShallowFBCSPNet	0.509	0.485	0.372	0.364	0.381	0.588	0.436	20 / 14 / 13 / 8
Mild vs Severe	Deep4Net	0.545	0.560	0.510	0.433	0.619	0.500	0.480	17 / 17 / 8 / 13

Majority-class baseline ACC (always predicting the larger group): None vs (Mild + Severe) 0.556 (69 / 124), (None + Mild) vs Severe 0.831 (103 / 124), Mild vs Severe 0.618 (34 / 55). Balanced ACC and AUC are 0.5 at chance level.

Selected hyper-parameters per outer fold

Each cell: learning rate, p = dropout, ep = final number of epochs (median of the 3 inner best epochs).

Task	Model	Outer fold 1	Outer fold 2	Outer fold 3	Outer fold 4
None vs (Mild + Severe)	EEGNet	1e-4, p 0.5, 1 ep	1e-3, p 0.5, 2 ep	1e-3, p 0.5, 1 ep	1e-3, p 0.5, 4 ep
None vs (Mild + Severe)	ShallowFBCSPNet	1e-4, p 0.5, 1 ep	1e-3, p 0.5, 1 ep	1e-4, p 0.5, 10 ep	1e-3, p 0.25, 3 ep
None vs (Mild + Severe)	Deep4Net	1e-3, p 0.5, 1 ep	1e-3, p 0.5, 2 ep	1e-3, p 0.5, 1 ep	1e-4, p 0.25, 2 ep
(None + Mild) vs Severe	EEGNet	1e-4, p 0.25, 1 ep	1e-3, p 0.5, 16 ep	1e-3, p 0.5, 1 ep	1e-3, p 0.5, 2 ep
(None + Mild) vs Severe	ShallowFBCSPNet	1e-3, p 0.5, 1 ep	1e-3, p 0.5, 2 ep	1e-3, p 0.5, 1 ep	1e-4, p 0.5, 5 ep
(None + Mild) vs Severe	Deep4Net	1e-4, p 0.25, 2 ep	1e-3, p 0.25, 2 ep	1e-4, p 0.25, 1 ep	1e-4, p 0.25, 2 ep
Mild vs Severe	EEGNet	1e-4, p 0.25, 3 ep	1e-3, p 0.25, 1 ep	1e-3, p 0.25, 1 ep	1e-3, p 0.5, 1 ep
Mild vs Severe	ShallowFBCSPNet	1e-3, p 0.5, 1 ep	1e-3, p 0.25, 1 ep	1e-4, p 0.5, 1 ep	1e-3, p 0.5, 1 ep
Mild vs Severe	Deep4Net	1e-4, p 0.25, 1 ep	1e-4, p 0.5, 1 ep	1e-3, p 0.25, 1 ep	1e-4, p 0.5, 5 ep

Experimental settings

All three models use the same protocol, grid and folds.

Category	Setting	Value
Input	Channels / window	F3, F4 / middle 300 s of each recording
	Sampling rate	128 Hz (from 1000 Hz, polyphase resampling)
	Segments	1-s windows, **75 % overlap** (0.25-s step), applied uniformly to all subjects: 1,197 per subject (148,428 in total)
	Normalisation	per-channel z-score, mean/std from the training subjects of each (inner or outer) training set
Outer CV	Scheme	stratified 4-fold, **split by subject**; outer-test subjects are never used for model selection
	Subjects (train / test)	93 / 31 (None vs (Mild + Severe), (None + Mild) vs Severe); 41–42 / 13–14 (Mild vs Severe)
Inner CV	Scheme	stratified 3-fold on the outer-training subjects, split by subject (2/3 train, 1/3 validation, rotated)
	Subjects (train / validation)	62 / 31; 27–28 / 13–14 (Mild vs Severe)
	Hyper-parameter grid	learning rate {1e-3, 1e-4} × dropout {0.25, 0.5} (4 configs)
	Best epoch per inner fold	lowest class-weighted validation loss; max 50 epochs, early stopping patience 10
	Config selection	highest mean subject-level validation balanced accuracy over the 3 inner folds (ties: lower mean validation loss)
	Final number of epochs	median of the selected config's 3 inner best epochs (rule fixed in advance)
Final model	Training	retrained from scratch on all outer-training subjects for the final number of epochs, no validation data
Training	Loss	cross-entropy weighted by inverse class frequency (training set)
	Optimiser / LR schedule	AdamW (lr from grid, weight decay 1e-4) / cosine annealing over 50 epochs (same trajectory in inner and final training)
	Batch size	128 segments
	Random seeds	42 (outer split), 42 + outer fold (inner split), 42 + 100 · outer + inner (inner runs), 42 + 100 · outer (final model)
Evaluation	Subject level	mean of the subject's 1,197 segment probabilities, threshold 0.5
	Pooling	one outer-test prediction per subject; metrics computed once on all subjects (124; 55 for Mild vs Severe)
	Metrics	ACC, balanced ACC, precision, recall (sensitivity), specificity, F1, ROC-AUC, confusion matrix

Subjects are split into folds. Splitting by segment would put 1-s pieces of the same recording in both training and test sets. The model could then recognise the person instead of their depression level, which inflates accuracy. With 75 % overlap this would be even worse, because neighbouring windows share 96 of their 128 samples.

Models

Model	Configuration	Params	Size (fp32)	FLOPs / 1-s segment
EEGNet (`EEGNetv4`)	braindecode (EEGNet-8,2; kernel 64 = 0.5 s at 128 Hz); dropout from grid	1,266	5.3 KB	0.31 M
ShallowFBCSPNet	braindecode defaults; dropout from grid	4,482	17.8 KB	0.42 M
Deep4Net	kernels 5, pools 2/2 (see below); dropout from grid	135,002	530 KB	3.64 M

The default Deep4Net (temporal kernel 10, max-pool 3/3 in four blocks, designed for 250 Hz / 4-s trials) cannot process a 128-sample input: the time axis shrinks below 1. Kernels and pools are scaled to roughly 128/250 of the original (kernel 5, pool 2 stride 2), giving time lengths 128 → 62 → 29 → 12 → 4. Filter counts and everything else are unchanged.

Environment

Lab server with 8 × NVIDIA Quadro M6000 (12 GB); Python 3.11.7, PyTorch 2.1.2 (CUDA 12.1), braindecode 0.8. The 468 training runs (432 inner + 36 final) took 52 min, with the 9 task × model jobs running in parallel. cuDNN was set to deterministic mode, and a re-run on the same GPU type reproduced the results exactly.